

메카트로닉스 (공) (MECH312-01)

학기	2026년 2학기
담당교수	송철
이수구분	교선평수
학점	3.0
강의시간	화10:30-12:00(E5 - 217), 목10:30-12:00(E5 - 217)
강의실	E5 513 (교수실)
소속	기초학부

교과개요

This course covers simultaneous understanding of the mechanical and electrical component characteristics. We will also learn design and analysis techniques of mechanical and electronic control systems, and construct and analyze real mechatronics systems using them. The contents of this course are as follows: (a) dynamic modeling and analysis on behavior of mechanical and electrical elements (b) motion control and analysis of the mechanical and electrical systems (c) robot mechatronics (d) measurement of physical quantities (acceleration, force, angle, torque etc.) and sensors via signal processing (e) artificial intelligence including machine/computer vision.

교과목표

Based on the theory, a simple mechatronic system will be selected and produced by each team. Each team will have more than two students. It requires us to conduct several presentations about plans, progress and final report throughout the whole semester.

학습윤리

DGIST Honor Code governs all work in this course (e.g., tests, papers, and homework assignments). Note that discussions among students are fine, but group solution of homework is not allowed. However, violations of this guidelines will not be tolerated and will be forwarded to the Honor Council. This policy also includes proper citation of sources and adherence to the anti-plagiarism policy. Also, a sample IEEE guidance is available at <https://www.ieee.org/content/dam/ieee-org/ieee/web/org/pubs/plagiarism.pps>

교과정책(방침)

□Prerequisite courses: Introduction to Control Engineering, 로봇전자공학. □Recommended courses: Introduction to Electrical Engineering (SE363). □For this course, the use of tools to analyze the dynamic characteristics and control characteristics of mechatronics systems (i.e., ANSYS, LS-DYNA, ADAMS, NI-LabVIEW, Matlab/Simulink, Arduino, C++ and etc.) is very helpful for achieving team project. □Since several S/W tool (e.g., NI LabVIEW) training sessions will be provided, it is necessary to learn while practicing tasks and projects while doing separate exercises. □All team members should actively be involved and speak with each own voice at several presentations about their team project.